FLOW ANALYSIS OF LAMINAR WALL JET

Under The Supervision of

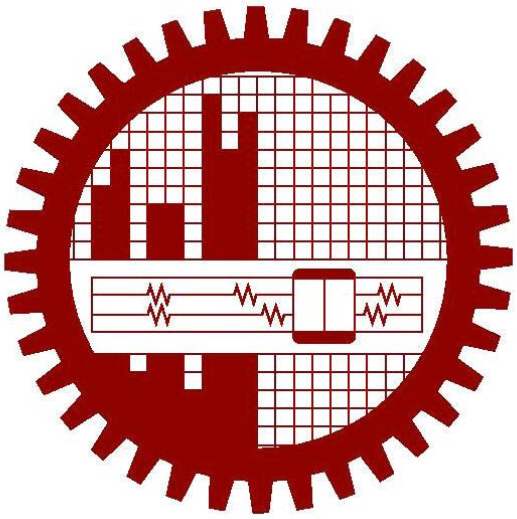
DR. MD. ABDUL AZIM

Professor

Department of Mechanical Engineering

Bangladesh University of Engineering and Technology (BUET)

Dhaka-1000.



CERTIFICATION

This is to certify that the work presented in this thesis is an outcome of the investigation carried out by the candidates under the close supervision of Dr. Md. Abdul Azim, Professor, Department of Mechanical Engineering, Bangladesh University of Engineering and Technology, Dhaka.

Signature of the Supervisor

DR. MD. ABDUL AZIM

Professor

Department of Mechanical Engineering

Bangladesh University of Engineering & Technology (BUET)

Dhaka -1000, Bangladesh

DECLARATION

It is hereby declared that this thesis or any part of this thesis has not been submitted elsewhere for the award of any degree or diploma or publication.

Candidates

_____

Ashraful Alam Limon (0910064)

_____

Shadmanur Rahman (0910098)

_____

Md. Najmul Alam Patwary (0910102)

ACKNOWLEDGEMENT

We are thankful to our almighty creature to whom everything belongs. We would like to express heartfelt gratitude to our supervisor Dr. Md. Abdul Azim, Professor, Department of Mechanical Engineering, BUET, Dhaka, for his encouragement to study thesis papers & journals, his patronizing guidance & untiring support throughout the thesis work. We are deeply indebted to him for his constructive advice & encouraging words, which have always been a constant source of inspiration.

ABSTRACT

This thesis has been performed for finding similarity solution of plane turbulent mixing layer. Considering the above situation continuity equation and momentum equation have been derived. Then, considering the momentum equation for turbulent fluid flow a third order ordinary differential equation has been derived using similarity transformations which are the governing equation. Finally numerical solution of the governing equation has been achieved by the improvisation of known boundary conditions into the initial boundary conditions. Here, MATLAB has been used to develop a computer program to solve the governing equation using fourth order Runge-Kutta method.

CONTENTS

Title		Page no.
ACKNOWLEDGEMENT	:	4
ABSTRACT	:	5
CONTENTS	:	6
NOMENCLATURE	:	7
1. INTRODUCTION	:	8
2. FORMULATION	:	10
3. RESULTS & DISCUSSION	:	11
4. CONCLUSION	:	16
REFERENCES	:	17
APPENDIX A	:	18
APPENDIX B	:	20
APPENDIX C	:	24
APPENDIX D	:	26
APPENDIX E	:	27

NOMENCLATURE

SYMBOL	MEANING
P	Pressure
u	Velocity component along X axis
v	Velocity component along Y axis
w	Velocity component along Z axis
ρ	Density of the fluid
μ	Absolute viscosity
ν	Kinematic viscosity
φ	Stream function
η	Non dimensional distance
f(η)	Non dimensional stream function

Introduction

In an actual environment the turbulent wall jets are dominant. While the laminar ones still attract many researchers owing to their numerous practical and applications.

A wall jet is formed when a fluid jet issues from an orifice and flows along a wall on one side while experiencing a free surface on other side. Thus the velocity distribution acquires the nature of a boundary layer near the wall but becomes that of a free jet at a large distance from the wall. The properties of a wall jet are relevant to many problems of fluid flow, boundary layer control, film cooling etc.

The theory of such wall jets, radial or plane, laminar or turbulent forms the subject of this paper.

Wall jet flow has many practical applications. Wall jet flow is commonly used in the windshield defroster system in automobiles. A jet of hot air is blown along the windshield to protect the surface from its surroundings. Introducing a recirculation by means of a step to enhance the heat transfer is an interesting study on this situation. The recirculation is used to shift the peak Nusselt number location in the downstream direction. Another example can be found from the cooling of electronic components. Steplike geometries are very often found in multiprocessor electronic components. For better performance, they are designed to maintain a constant low temperature. Wall jets are used to cool such hot surfaces. The knowledge of recirculation eddy size is technically important from the design point of view in connection with Re and the step length and height. It is a common belief that the peak Nusselt number will occur near the reattachment location. So, it is very important to find the reattachment location in connection with Re and geometry. Wall jet flow has several similar applications like a refrigerated air curtain, the paper industry, etc. The understanding of the flow behavior of the wall jet is important in engineering practices. If attachment is not desired for more mixing, a knowledge of calculation and design is required to prevent it. For a case where attachment is desirable, study of the involved variables is required so that a precise location of the attachment point and the containment of the flow can be established.

Analytical solutions of the wall jet are available, based on the self-similarity of the velocity field. However, these solutions are valid only far away from the jet inlet and, in most applications, the near-field development holds the key to important features of the jet flow and recirculation. Therefore, the near-field development of a wall jet has been the subject of a lot of research in recent years.

The flow emanating from a two-dimensional 2D plane wall jet is shown in Figure where the main features and regions of interest are depicted.

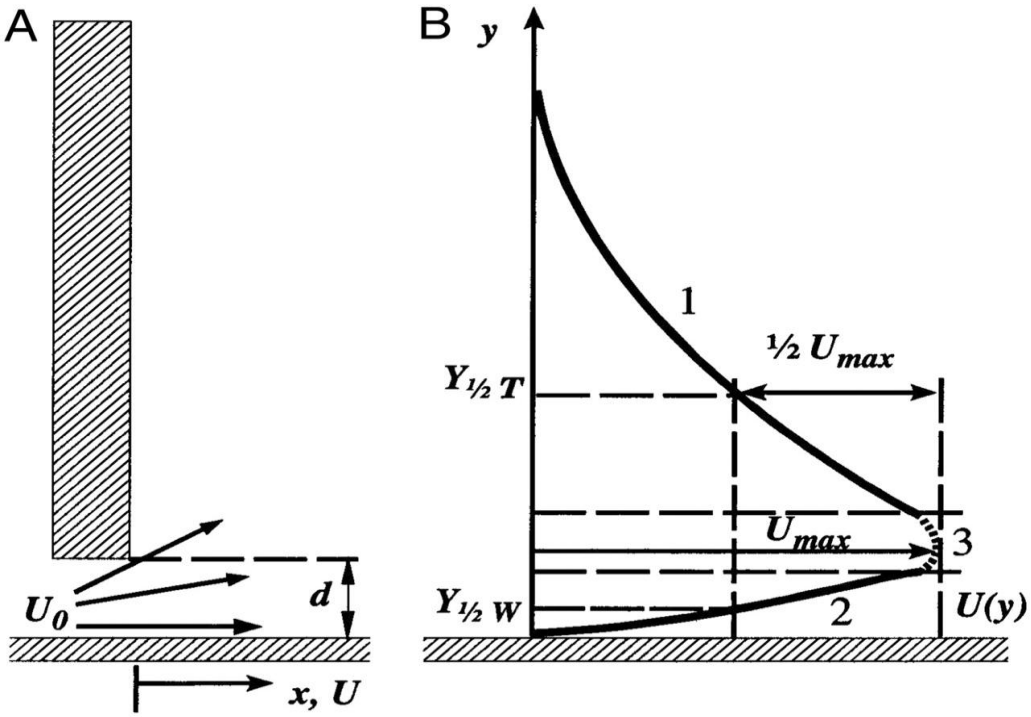


Figure 1: Schematic diagram of plane wall jet under step

FORMULATION

Assuming the fluid is incompressible and the flow is steady and no pressure variation along perpendicular direction of flat plate, the governing equations for fluid motion are given by-

Continuity Equation for turbulent case:

$$\frac{\delta u}{\delta x} + \frac{\delta v}{\delta y} = 0$$

Momentum Equation for turbulent case:

$$u\frac{\delta u}{\delta x} + v\frac{\delta u}{\delta y} = \gamma\frac{\delta^2 u}{\delta y^2}$$

Using similarity transformation the third ordinary differential equation has been derived from the partial differential momentum equation.

Similarity transformations are

$$u = x^{-1/2} f'$$

$$v = \frac{1}{4}\gamma^{1/2}\, x^{-3/4} (3\eta f' - f)$$

The governing ordinary differential equation,

$$4f''' + ff'' + 2(f')^2 = 0$$

Boundary conditions:

$$f(0)=0$$

$$f'(0)=0$$

$$f'(\infty) = 0$$

But these boundary conditions cannot be used to solve the above governing differential equation using 4th order Runge-Kutta method. Because, in Runge-Kutta method all the boundary conditions should be at initial value of independent variable. So, we need to improvise the given boundary conditions (at $\eta = -\infty, +\infty$) to initial conditions (at $\eta = 0$).

Improvisation of known boundary conditions

The initial conditions are required to solve a third order ordinary differential equation, the value of $f''(0)$ is necessary. But there is no such condition that the value $f''(0)$ can be found out. For this reason solution has been found out by iterating the computer program for various assumed values of $f'(0)$.

The solution is shown in the results and discussions section.

RESULTS AND DISCUSSION

The obtained results from the numerical solutions is shown below-

Flow properties and flow geometry

For x=1,

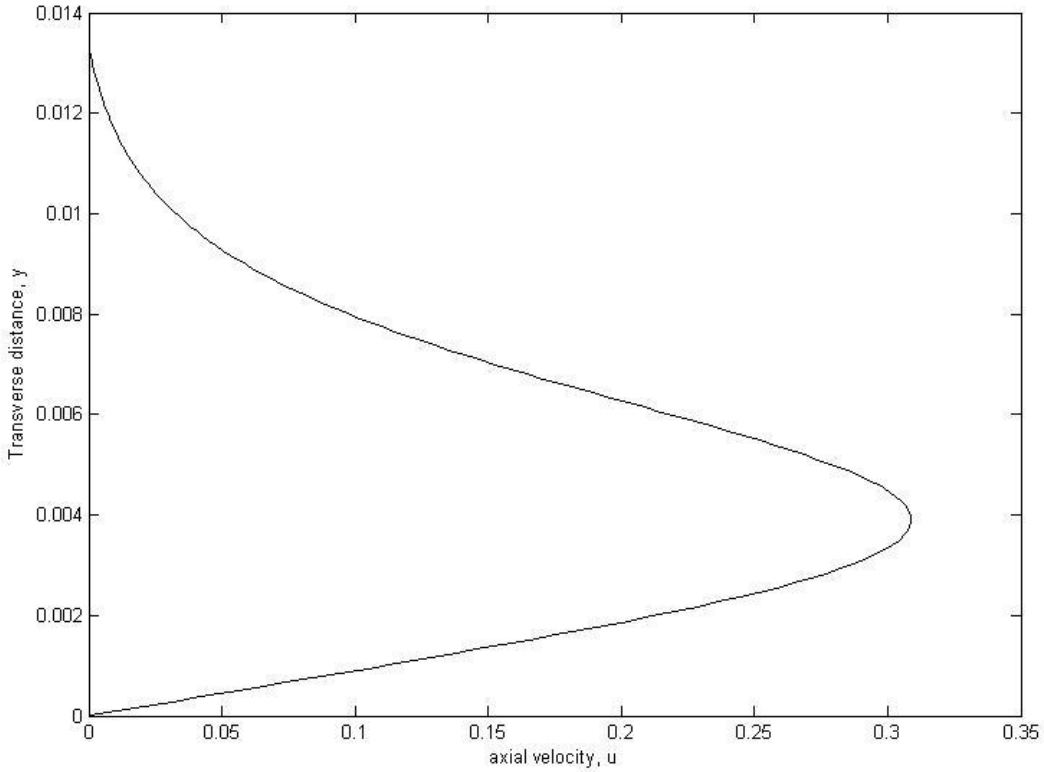


Figure 2: Axial velocity against transverse distance.

Form the equation of axial velocity u it can be seen that u is proportional to f'(η). So the above graph indicates the profile of axial velocity u with respect to η.

From the graph we can see that the values of axial velocity increases up to certain limit then it became zero again.

For x= 2,

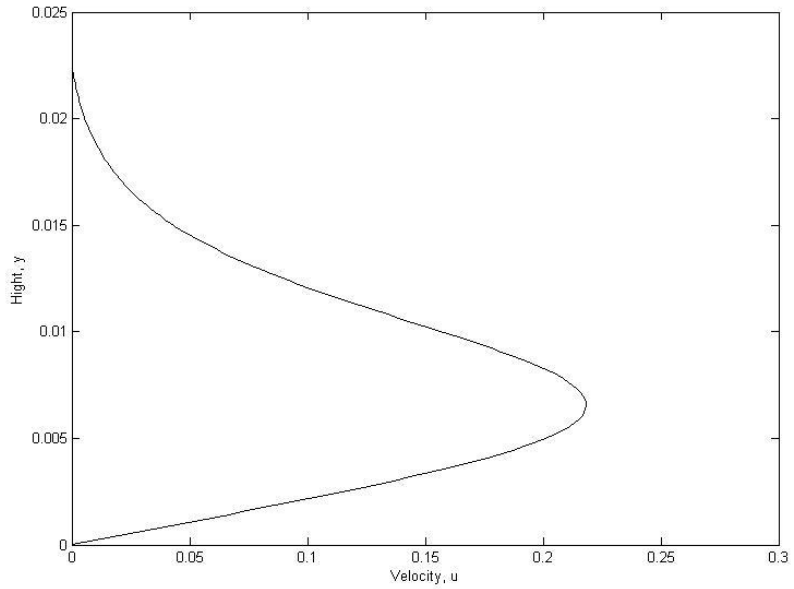


Figure 3: Axial velocity against transverse distance.

For x=3,

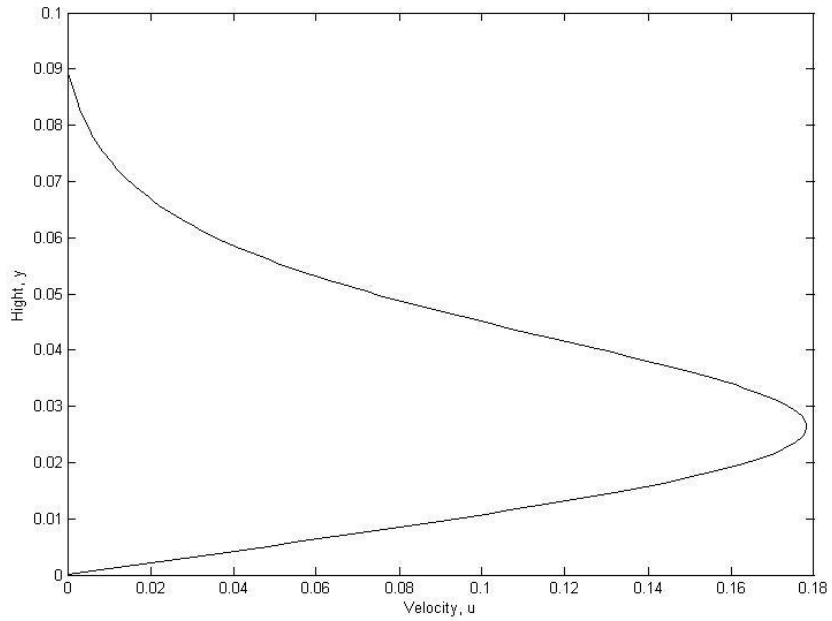


Figure 4: Axial velocity against transverse distance.

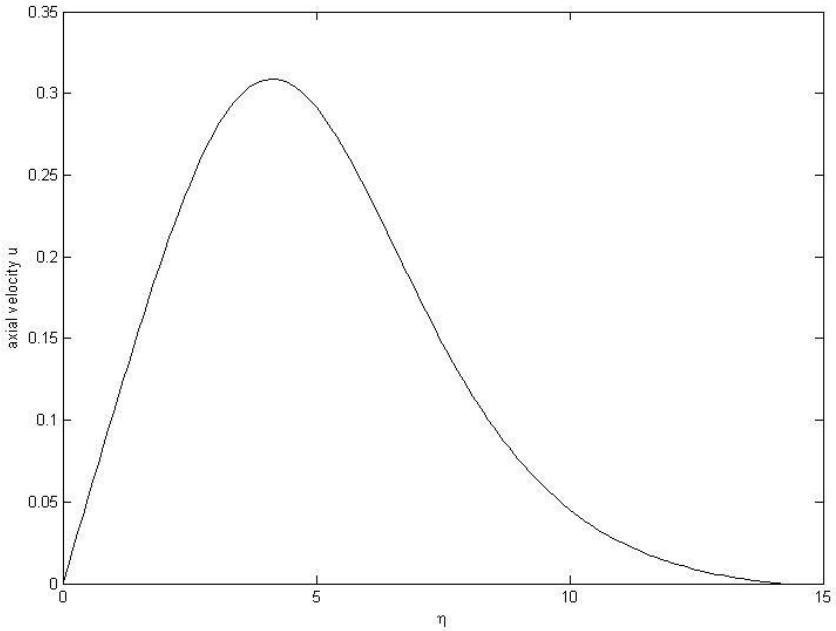
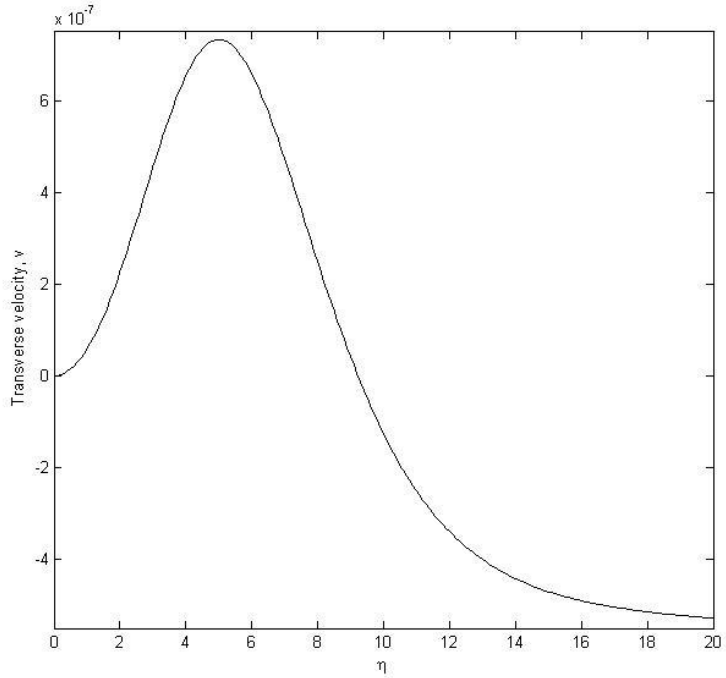


Figure 5: Axial velocity with respect to η

Form the graph for different value of x, it is seen that as value of is increased maximum axial velocity decreased but the free surface area is increasing where perpendicular velocity is zero.

Transverse Velocity:



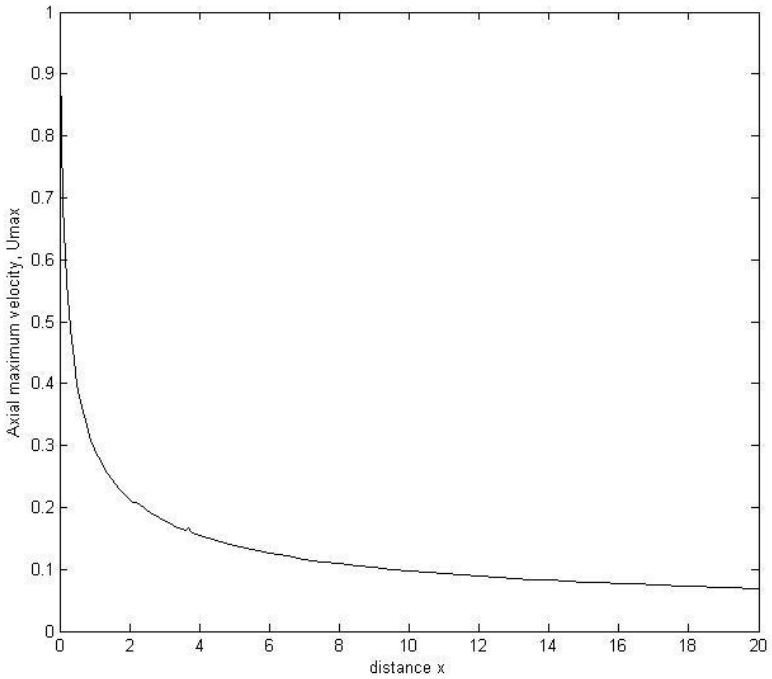


Figure 7: Axial maximum velocity against axial distance

Stream function

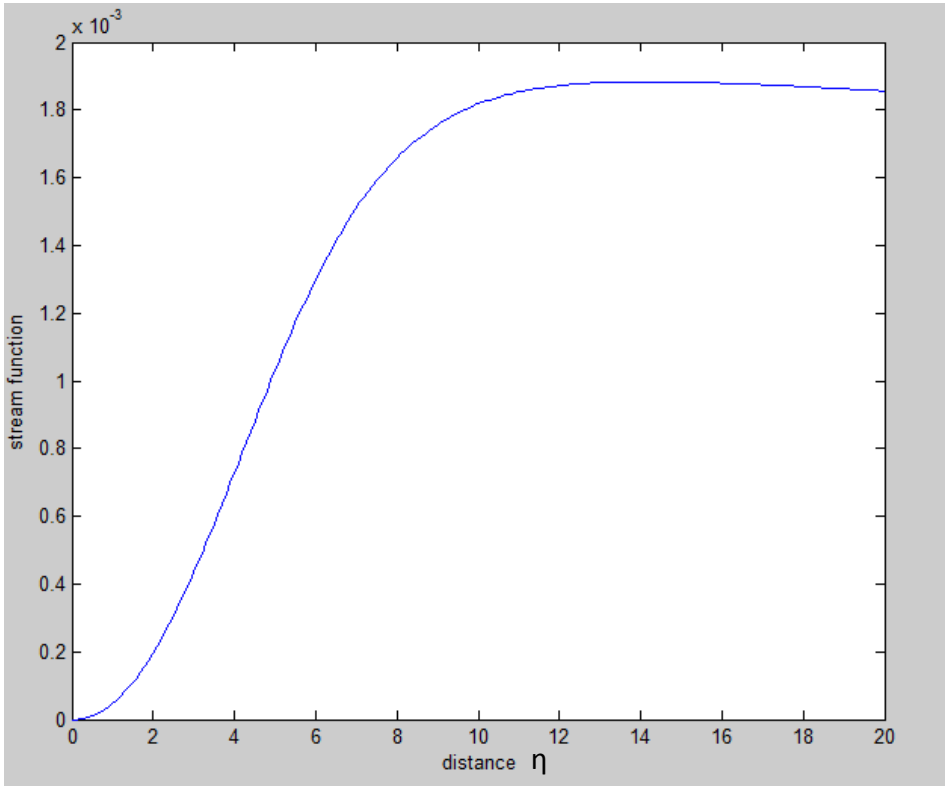


Figure 8: Stream function vs non-dimensional distance

As we see from this graph value of stream function is increasing for a certain value of η.

Shear stress

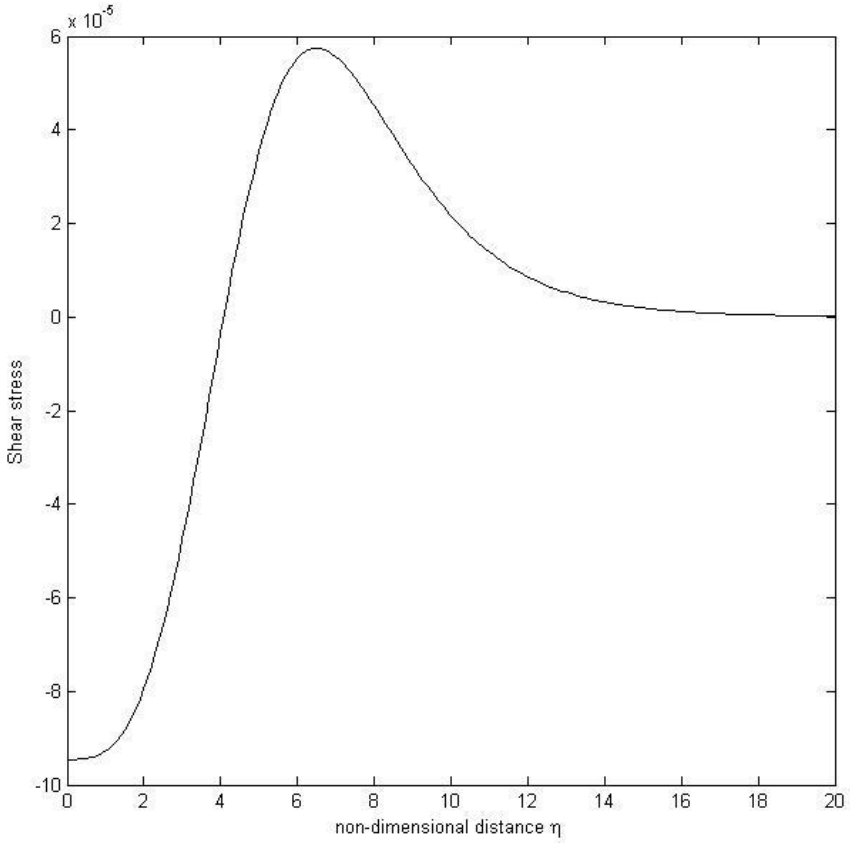


Figure 9: Shear stress against non-dimensional distance

CONCLUSION

Fluid flows are governed by the Navier-Stokes equations, for which there are no particularly useful analytical solutions. The equation is therefore usually solved by numerical procedures over a domain discretized into finite elements or volumes. The numerical solutions do not provide a global view of the problem, because in contrast to the general character of analytical solutions each computed case is valid only for a particular set of parameters and boundary condition values, not showing a relation with other cases. This is acceptable for solving particular engineering task, but not for general investigation.

We have neglected the pressure gradient by uniform pressure assumption and viscous shear forces being small in deriving the momentum equation.

REFERENCES

1. Dr. Hermann Schlinchting, *Boundary-Layer Theory, 1979, McGraw Hill, New York.*

2. Munson R., *Fundamentals of Fluid Mechanics, Sixth edition, 2010, Wiley*

3. Frank M White, *Fluid Mechanics*, 2008, *McGraw Hill, New Delhi.*

4. Blagurasamy E., *Numerical Methods*, fourth edition, 1995, Tata *McGraw Hill, New Delhi.*
5. Pratap Rudra, *Geetting Started with MATLAB 7/ A Quick Introduction for Scientists and Engineers,* first edition, 2006, Oxford University Press.

Appendix-A

The differential equation of mass conservation:-

Here we choose an infinitesimal fixed control volume (dx, dy, dz). The flow through each side of the element is approximately one dimensional, and so the appropriate mass conservation relation to use here is:

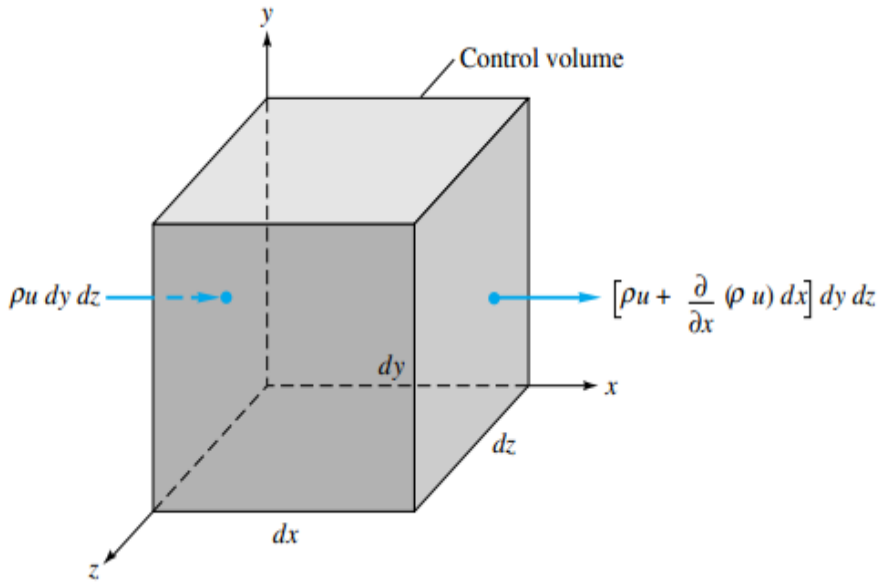


Figure 10: Elemental Cartesian fixed control volume showing the inlet and outlet mass flow on the x face.

$$\int_{cv} \frac{\delta \rho}{dt} dV + \sum_{i} (\rho i \; Ai \; Vi) \; out - \sum (Pi \; Vi \; Ai) \; in = 0 \; \ldots\ldots\ldots\ldots . \text{(i)}$$

As the volume is so small the volume integral is simply reduced to-

$$\int_{cv} \frac{\delta \rho}{dt} dV \approx \frac{\delta \rho}{dt} dx dy dz$$

Face	Inlet mass flow	outlet mass flow
x	$\rho u \; dy \; dz$	$\left[\rho u + \frac{\delta}{\delta x}(\rho u) dx\right] dy \; dz$

y	$\rho v\ dx\ dt$	$\left[\rho uv + \frac{\delta}{\delta x}(\rho v)dy\right] dx\ dz$
z	$\rho w\ dx\ dy$	$\left[\rho uv + \frac{\delta}{\delta x}(\rho w)dz\right] dx\ dy$

So e equation (i) reduced to,

$$\frac{\delta \rho}{\delta t}.dx\ dy\ dz + \frac{\delta}{\delta x}(\rho u)dx\ dy\ dz + \frac{\delta}{\delta y}(\rho v)dx\ dy\ dz + \frac{\delta}{\delta t}(\rho w)dx\ dy\ dz = 0$$

$$= \frac{\delta p}{\delta t} + \frac{\delta}{\delta x}(\rho u) + \frac{\delta}{\delta y}(\rho v) + \frac{\delta}{\delta t}(\rho w) = 0$$

$$= \frac{\delta \rho}{\delta t} + \nabla.(\rho u) = 0 \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(ii)}$$

This is the compact from of continuity equation.

For 2-D, & steady flow we get,

$$\frac{\delta u}{\delta x} + \frac{\delta v}{\delta y} = 0$$

Appendix-B

The appropriate from of linear momentum relation is,

$$\sum F = \frac{\delta}{\delta t}\left(\int_{cv} v\rho dv\right) + \sum (mi\ vi)\ out - \sum (mi\ vi)\ in \ \ldots\ldots\ldots (iii)$$

As the element is so small that the volume integral simply reduces to-

$$\frac{\delta}{\delta t}(v\rho\ dv) \approx \frac{\delta}{\delta t}(\rho v) dx\ dy\ dz \ \ldots\ldots\ldots\ldots\ldots (iv)$$

Face	Inlet mass flow	outlet mass flow
x	$\rho uV\ dy\ dz$	$\left[\rho uV + \frac{\delta}{\delta x}(\rho uV)dx\right] dy\ dz$
y	$\rho vV\ dx\ dt$	$\left[\rho vV + \frac{\delta}{\delta x}(\rho vV)dy\right] dx\ dz$
z	$\rho wV\ dx\ dy$	$\left[\rho wV + \frac{\delta}{\delta x}(\rho wV)dz\right] dx\ dy$

From equation (iv) & this turns we found in equation (iii)

$$\sum F = dx\ dy\ dz\ \left[\frac{\delta}{\delta t}(\rho v) + \frac{\delta}{\delta x}(\rho uv) + \frac{\delta}{\delta y}(\rho uv) + \frac{\delta}{\delta t}(\rho wv)\right]$$

$$=\left[dx\ dy\ dz\ \left[V\frac{\delta p}{\delta t} + \nabla.(\rho V)\right] + \rho\left(\frac{\delta V}{\delta t} + u\frac{\delta}{\delta x}^{V} + V\frac{\delta V}{\delta y} + w\frac{\delta V}{\delta t}\right)\right]$$

$$= dx\ dy\ dz\ \times \frac{dV}{dt}[From\ chain\ rule\ of\ Acceleration]$$

$$\therefore \sum F = \rho\frac{dV}{dt} dx\ dy\ dz$$

Here $\sum F$ *is two types*

i) Body force

ii) Surface force

$$\sum F = \sum F_{body} + \sum F_{surface.}$$

For body force we consider only graviton force

$\therefore dFgrav = \rho g\, dx\, dy\, dz$

Surface force:-

i) Shear stress,τ

ii) Normal force, σ

Shear stress:-

$$\therefore \sigma_{ij} = \begin{vmatrix} -\rho + \tau_{xx} & \tau_{yz} & \tau_{zx} \\ \tau_{xy} & -\rho + \tau_{yy} & \tau_{zy} \\ \tau_{xt} & \tau_{yz} & -\rho + \tau_{zz} \end{vmatrix}$$

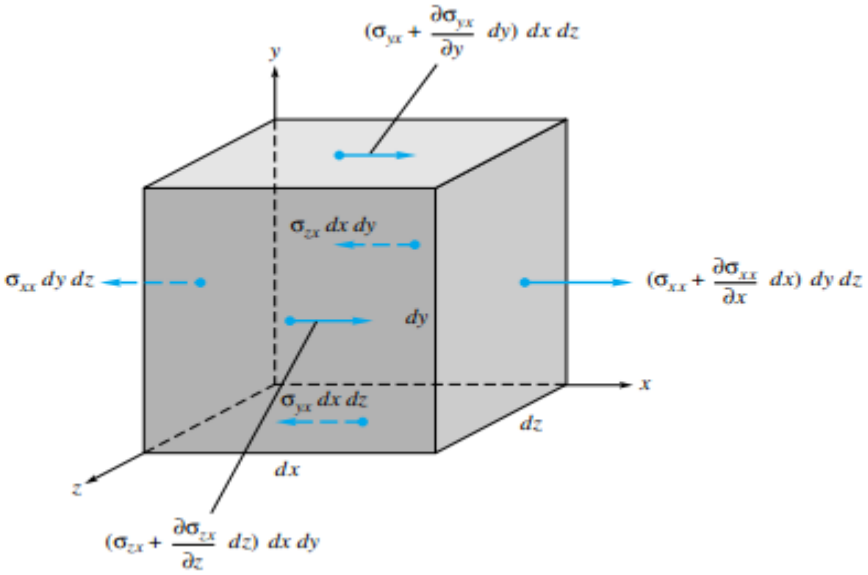


Figure 11: Elemental Cartesian fixed control volume showing the surface force in the x-direction

$\sigma_{ij} = stress\ in\ J\ direction\ on\ the\ face\ normal\ to\ i\ axis.$

Elemental Cartesian fixed control volume showing the surface force in the x- direction only.

$$dF_x, surf = \left[\frac{\mathcal{S}}{\mathcal{S}x}(\sigma_{xx}) + \frac{\mathcal{S}}{\mathcal{S}y}(\sigma_{yz}) + \frac{\mathcal{S}}{\mathcal{S}z}(\sigma_{zx})\right] dx\, dy\, dz \ \ldots..(v)$$

$$\therefore \frac{dFx}{dV} = -\frac{d\rho}{\mathcal{S}x} + \frac{\mathcal{S}}{\mathcal{S}x}(\tau_{xx}) + \frac{\mathcal{S}}{\mathcal{S}y}(\tau_{yx}) + \frac{\mathcal{S}}{\mathcal{S}z}(\tau_{zx}) \ldots..(vi)$$

Here,

dV =dx*dy*dz

$\therefore from\ eq\ (v)\ \&\ (vi) for\ 3-D$

$$\left(\frac{\mathcal{S}F}{dv}\right)_{xvrt} = -\nabla p + \left(\frac{dF}{dv}\right)_{viscous.}$$

Viscous force,

$$\left(\frac{dF}{dV}\right)_{viscous} = i\left(\frac{\mathcal{S}\tau_{xx}}{\mathcal{S}x} + \frac{\mathcal{S}yx}{\mathcal{S}y} + \frac{\mathcal{S}\tau_{zx}}{\mathcal{S}t}\right) + j\left(\frac{\mathcal{S}\tau_{xy}}{\mathcal{S}x} + \frac{\mathcal{S}yy}{\mathcal{S}y} + \frac{\mathcal{S}\tau_{zy}}{\mathcal{S}t}\right) + j\left(\frac{\mathcal{S}\tau_{xz}}{\mathcal{S}x} + \frac{\mathcal{S}yt}{\mathcal{S}y} + \frac{\mathcal{S}\tau_{zz}}{\mathcal{S}t}\right)$$

$$\left(\frac{\mathcal{S}F}{dV}\right)_{viscous} = \nabla.\tau_{ij}$$

Here $\tau_{ij} = \begin{bmatrix}\tau_{xx} & \tau_{xy} & \tau_{zx}\\ \tau_{xy} & \tau_{yy} & \tau_{zy}\\ \tau_{xz} & \tau_{yz} & \tau_{zz}\end{bmatrix}$

From

$$\therefore \rho g - \nabla\rho + +\nabla.\tau_{ij} = \rho\frac{dV}{dt} \ldots\ldots\ldots (vii)$$

for 2-D & steady flow this equation reduced to,

$$\rho g_x - \frac{\mathcal{S}\rho}{\mathcal{S}x} + \frac{\mathcal{S}\tau_{xx}}{\mathcal{S}x} + \frac{\mathcal{S}\tau_{yx}}{\mathcal{S}y} = \rho\left(u\frac{\mathcal{S}u}{\mathcal{S}x} + \frac{v\mathcal{S}u}{\mathcal{S}y}\right) \ldots\ldots\ldots (viii)$$

$$\rho g_y - \frac{\mathcal{S}\rho}{\mathcal{S}y} + \frac{\mathcal{S}\tau_{xx}}{\mathcal{S}x} + \frac{\mathcal{S}\tau_{yx}}{\mathcal{S}y} = \rho\left(u\frac{\mathcal{S}v}{\mathcal{S}x} + \frac{v\mathcal{S}v}{\mathcal{S}y}\right) \ldots\ldots\ldots (ix)$$

For viscid flow, incompressible flow, 2D steady flow,

$$\tau_{xx} = 2\mu\frac{\mathcal{S}u}{\mathcal{S}x}, \tau_{yy} = 2\mu\frac{\mathcal{S}v}{dy}$$

$$\tau_{xy} = \tau_{yx} = \mu\left(\frac{\mathcal{S}u}{\mathcal{S}y} + \frac{\mathcal{S}v}{\mathcal{S}x}\right)$$

From equation (viii) & (ix) we get,

$$\rho g_x - \frac{\delta \rho}{\delta x} + \mu \left(\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} \right) = \rho \frac{\delta u}{dt}$$

$$\rho g_j - \frac{\delta \rho}{\delta x} + \mu \left(\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} \right) = \rho \frac{\delta v}{dt}$$

These are the in-compressible flow Navier- stokes eqn.

If pressure force is zero and

$$\frac{\delta}{\delta x} < \frac{\delta}{\delta y}, \frac{\delta^2}{\delta x^2} \ll \frac{\delta^2}{\delta y^2}$$

and neglcting gravitional force in x & y *direction.*

$$\gamma \frac{\delta^2 u}{\delta y^2} = u \frac{\delta u}{\delta x} + v \frac{\delta u}{\delta y}$$

$$\gamma \frac{\delta^2 v}{\delta y^2} = u \frac{\delta v}{\delta x} + v \frac{\delta v}{\delta y}$$

y-momentum or v momentum or cross stream momentum eqn negligible compared to u-momentum eqn.

So, we get,

$$u \frac{\delta u}{\delta x} + v \frac{\delta u}{\delta y} = \gamma \frac{\delta^2 u}{\delta y^2}$$

Appendix -C

Development of Ordinary Differential Equation with the help of similarity transformation:

As we know the similarity variables

$$\varphi = \gamma^{1/2} x^{1/4} \, f(\eta) \qquad \eta = \gamma^{-1/2} \, x^{-3/4} \, y$$

From stream function,

$$u = \frac{\partial \varphi}{\partial y} \, , \qquad v = -\frac{\partial \varphi}{\partial x}$$

Using von-mesh transformation chain rule,

$$u = \frac{\partial \varphi}{\partial y} = \frac{\partial \varphi}{\partial \eta} . \frac{\partial \eta}{\partial y} + \frac{\partial \varphi}{\partial x} . \frac{\partial x}{\partial y}$$

$$= \frac{\partial \varphi}{\partial \eta} . \frac{\partial \eta}{\partial y} \quad \ldots\ldots\ldots\ldots (1)$$

$$v = -\frac{\partial \varphi}{\partial x} = -[\frac{\partial \varphi}{\partial \eta} . \frac{\partial \eta}{\partial x} + \frac{\partial \varphi}{\partial x}]$$

In $\eta - x$ plane we get,

$$\frac{\partial \varphi}{\partial x} = \frac{1}{4} \gamma^{1/2} x^{-3/4} f$$

$$\frac{\partial \eta}{\partial x} = -\frac{3}{4} - y x^{-7/4} \gamma^{-1/2}$$

$$\frac{\partial \eta}{\partial y} = \gamma^{-\frac{1}{2}} x^{-\frac{3}{4}} \quad \text{And ,}$$

$$\frac{\partial \varphi}{\partial \eta} = \gamma^{1/2} x^{1/4} \, f'$$

From equation (1),

$$u = x^{-1/2} f'$$

And from equation (2) the cross stream velocity,

$$v = -[\frac{\partial \varphi}{\partial x}.\frac{\partial \eta}{\partial x} + \frac{\partial \varphi}{\partial x}]$$

$$= \frac{1}{4}\gamma^{1/2} x^{-3/4} (3\eta f' - f)$$

Again,

$$\frac{\partial u}{\partial y} = \gamma^{-1/2} x^{-5/4} f''$$

$$\frac{\partial u}{\partial x} = \frac{-3}{4} y\, x^{-3/4} \gamma^{-1/2} f''$$

$$\frac{\partial v}{\partial y} = \frac{3}{4} x^{-3/4} f'$$

$$\frac{\partial^2 u}{\partial y^2} = f'''/\gamma x^2$$

From appendix-B we get the momentum equation in the following partial differential form-

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \gamma\frac{\partial^2 u}{\partial y^2}$$

$$x^{-\frac{1}{2}} f' \left[-\frac{3}{4} x^{-\frac{3}{4}} \gamma^{-\frac{1}{2}} y f'' - \frac{1}{2} x^{-\frac{3}{4}} f'\right] + \frac{1}{4}\gamma^{\frac{1}{2}} x^{-\frac{3}{4}} [3\eta f' - f] * \gamma^{-\frac{1}{2}} x^{-\frac{5}{4}} f'' = \frac{f'''}{x^2}$$

$$4f''' + ff'' + 2(f')^2 = 0$$

This is the ordinary differential equation transformed from the partial differential momentum equation

Appendix –D

The Algorithm For 4th order Runge –Kutta Method:

The algorithm of 4th order Runge-Kutta method for general solution of a 2nd order differential equation is shown below

$$y_{n+1} = y_n + (k_1 + k_2 + k_3 + k_4)/6$$

Where, $k_1 = hf(x_n, y_n)$

$$k_{2=} hf\left(x_n + \frac{h}{2}, y_n + \frac{hk_1}{2}\right)$$

$$k_{3=} hf\left(x_n + \frac{h}{2}, y_n + \frac{hk_2}{2}\right)$$

$$k_{4=} hf(x_n + h, y_n + hk_3)$$

For n=0, 1, 2, 3…….. given that y=y_0 when $x = x_0$ and for a step size of h

If there is a third order differential equation, then this equation is reduced to three first order equations. Then, these three first order equations are solved simultaneously using 4th order Runge-kutta method

Appendix –E

Three first order equations-

```
function f=f1(x, y, v, w)
f=v;
end
function f=f2(x, y, v, w)
f=w;
end
function f=f3(x, y, v, w)
f=-(2*v^2+y*w)/4;
end
```

Program for 4th order Runge-Kutta method

```
clear all;
clc;
h=0.1;
n=200;
x0=0;
y0=0;
v0=0;
w0=0.1;
x = linspace (0, 0, n+1);
y = linspace (0, 0, n+1);
v = linspace (0, 0, n+1);
w = linspace (0, 0, n+1);

x(1)=x0;y(1)=y0;v(1)=v0;w(1)=w0;

clc;
format long g
for i=1:n
i;
k1=f1(x(i),y(i),v(i),w(i));
l1=f2(x(i),y(i),v(i),w(i));
m1=f3(x(i),y(i),v(i),w(i));

k2=f1(x(i)+h/2,y(i)+k1*h/2,v(i)+k1*h/2,w(i)+k1*h/2);
l2=f2(x(i)+h/2,y(i)+l1*h/2,v(i)+l1*h/2,w(i)+l1*h/2);
m2=f3(x(i)+h/2,y(i)+m1*h/2,v(i)+m1*h/2,w(i)+m1*h/2);

k3=f1(x(i)+h/2,y(i)+k2*h/2,v(i)+k2*h/2,w(i)+k2*h/2);
l3=f2(x(i)+h/2,y(i)+l2*h/2,v(i)+l2*h/2,w(i)+l2*h/2);
m3=f3(x(i)+h/2,y(i)+m2*h/2,v(i)+m2*h/2,w(i)+m2*h/2);

k4=f1(x(i)+h,y(i)+k3*h,v(i)+k3*h,w(i)+k3*h);
l4=f2(x(i)+h,y(i)+l3*h,v(i)+l3*h,w(i)+l3*h);
m4=f3(x(i)+h,y(i)+m3*h,v(i)+m3*h,w(i)+m3*h);

y(i+1)=y(i) +(k1+2*k2+2*k3+k4)*h/6;
v(i+1)=v(i) +(l1+2*l2+2*l3+l4)*h/6;
w(i+1)=w(i) +(m1+2*m2+2*m3+m4)*h/6;
x(i+1)=x(i)+h;

end
format long g
plot(x,y,'.',x,v,'--',x,w,'black');

u=x;
return;

end
```